

TrES-3b

R-0444

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_250726 · created 2026-07-19T14:51:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460882.7255 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.033
Transit depth (Rp/Rs)^2	3.59 +/- 1.25 [%]
Orbital Inclination (inc)	81.43 +/- 0.92
Airmass coefficient 1 (a1)	1.013 +/- 0.012
Airmass coefficient 2 (a2)	-0.014 +/- 0.064
Scatter in the residuals of the lightcurve fit is	1.21 %
Best Comparison Star	None
Optimal Aperture	3.57
Optimal Annulus	12.18
Transit Duration (day)	0.052 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.033 vs 0.16309 (deviation 0.7σ)
Duration fit vs published	0.052 d vs 0.0567 d (deviation 0.33σ)
Mid-time O–C	-4.8 min
Depth SNR	4.9
Residual scatter	1.21 %

Light curve

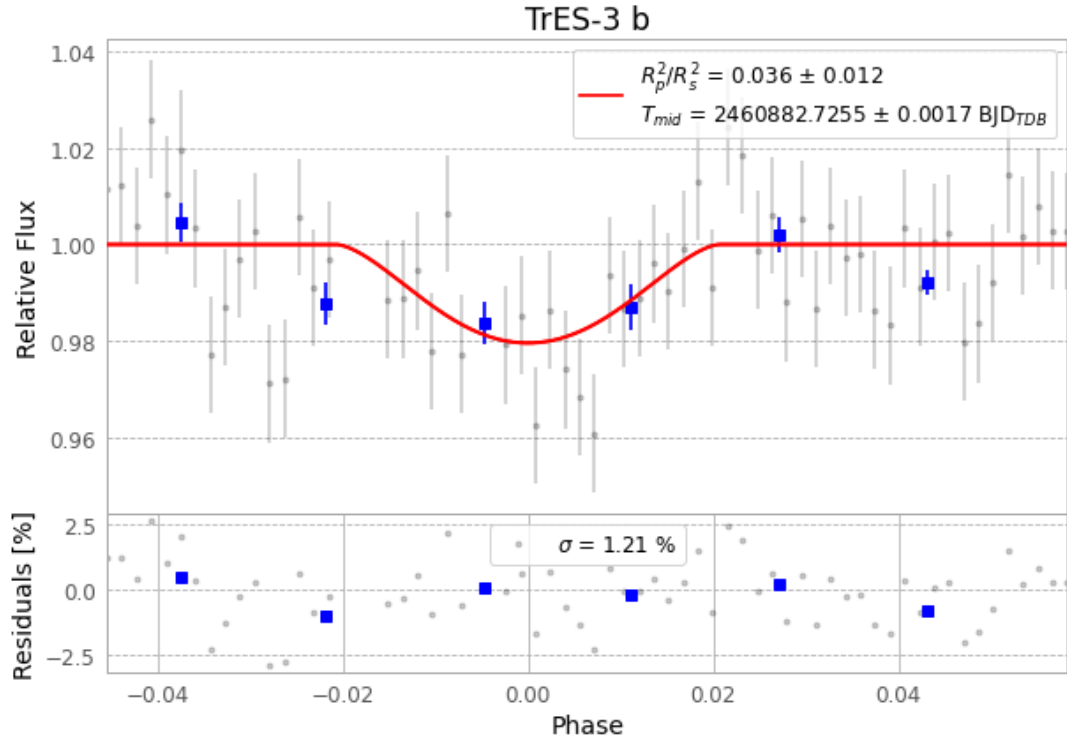


Figure 1 — FinalLightCurve_TrES-3 b_2025-07-25.png (SHA-256 9d69b6b56f381f18...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 229], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d5fa1a5c7b85a63a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

66 FITS frames (43 MB), TRES-3250726035416.FITS → TRES-3250726070916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250726.log (671b89186549...), FinalLightCurve_TrES-3 b_2025-07-25.png (9d69b6b56f38...), FinalLightCurve_TrES-3 b_2025-07-25.csv (1df96e436135...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 14:51Z.